

\$29 billion, India still lacks global brands. “We killed our brands in the 90s. How do we bring them back?” he challenged, pointing to India’s 500,000 annual engineering graduates as a resource that must now be channelled into building for the world and India itself.

Kishor Narang responded with a harsh statement: that **most of our engineers lack the competence to innovate and design. However, they are not to be blamed because our engineering institutes do not have faculty who can teach them design.** They are very good at research, but they themselves have never done a single product design because that is all they have learnt. They have never worked in the industry designing end-to-end products and/or systems. They are not familiar with the compliance standards that the products must comply with before being supplied to any market.

Even software standards like MISRA C and ISO 26262 are not taught in any software courses. MISRA (Motor Industry Software Reliability Association) C is the safe coding standard, and the standard ISO 26262 is an adaptation of the functional safety standard IEC 61508 for automotive electric/electronic systems. ISO 26262 defines functional safety for automotive equipment applicable throughout the lifecycle of all automotive electronic and electrical safety-related systems.

He further emphasised the criticality of building indigenous design capabilities in the strategic electronics domain: defence and aerospace, due to changing geopolitical scenarios.

India has started making great strides in manufacturing, but how about designing?

While India’s electronics manufacturing capabilities have soared, expected to hit \$150 billion this year, it can be unanimously agreed that

manufacturing is necessary but not sufficient. According to Josh Foulger, president of the electronics division of ZETWERK, the next five years as EMS 2.0, with the government targeting \$500 billion, can be a reasonable goal if everyone rows in the same direction. While many companies have proven the impact of government initiatives, there is an urgent need to shift from GCCs to true product innovation. Design hubs like Bengaluru must pivot from task-based execution to product ownership. “Qualify your product, validate it, and take it to mass production,” he urged.

And the biggest bottleneck? Components. India still imports critical components, such as PCBs and displays. Each component category requires its ecosystem. “We have checked the EMS 1.0 box. Now we have got to do India-plus-demand,” Josh said.

Millions of engineers, but where is the innovation?

The country has built factories, but what about the brains behind the products? Industry trailblazers often highlight a foundational issue: education. “We do not have an education system, we have an exam system,” stated Jaswinder. Bright engineering graduates from top schools often take two years to become productive. Engineering students are trained to pass tests, rather than to solve real-world problems or design products from end to end. Project-based learning can only be an alternative for big ESDM dreams.

“Our education system is more



Josh Foulger, President, Electronics, Zetwerk Manufacturing

qualification-oriented than knowledge-oriented,” noted Kishor. Though many institutions claim to encourage projects, a reboot is needed of how college projects are conceived; students should be designing hardware, coding from scratch, and working on micro-drones, power electronics, and real-world applications.

While industry-university collaboration is improving through government-backed incubation programmes, the pace of change is painstakingly slow.

“The time it takes to bridge this gap is still too long. Engineers struggle to understand compliance, standards, and validation. They are taught how to build a product, but not how to make it usable at scale,” said Shankar.

Is India’s strength as a chip designer a flawed notion, then? Jaswinder challenged, “Most of our engineers are task executors, not true chip designers.”

Reclaiming India’s large pool of talent

India leads the world in smartwatch usage and 5G rollout, yet contributes little to their design or IP. While government initiatives, such as the